

2(a)(i)	any two from: 1 (data is) continuous ; 2 (data is) normally distributed ; 3 standard deviations are (approximately) the same ; 4 fewer than 30, values / birds / AW (for each song type) ;	2 ignore comparing / difference, between (two) means (in question stem) 1 ignore quantitative data R discrete / discontinuous / categoric / nominal / qualitative, data 4 A 11 birds of song type A and 13 birds of song type B
2(a)(ii)	there is no, difference, in the (mean) bill length, between, (male) <i>G. conirostris</i> (with song) type A, and (male <i>G. conirostris</i> with song) type B ;	1 allow birds / finches / males, as AW for <i>G. conirostris</i> allow beaks as AW for bills ignore significant ignore (mean) mass ignore (mean) bill, depth / width R no, correlation / relationship / association R female (<i>G. conirostris</i>)
2(a)(iii)	22 ;	1 $11 + 13 - 2 = 22$

Question	Answer	Marks	Guidance
2(a)(iv)	1 calculated (t-) value / 3.768, is greater than, 2.074 (critical value at $df = 22$ and $p = 0.05$) ; ora 2 null hypothesis / H_0, is rejected ; 3 there is a <u>significant difference</u> ;	3	check Table 2.2 and answer for 2aiii ecf all mp for incorrect df / NR in 2aiii 1 ignore t-value unqualified (could be the calculated or critical values) need to state correct critical value (for their df) allow correct critical value from table (circled / underlined etc) mp 2 & 3 ecf incorrect critical value to max 2 2 R null hypothesis is accepted 3 R correlation / relationship / association 3 A greater than 95% probability that the <u>difference</u> (between the means) is not due to chance / is significant or less than 5% probability that the <u>difference</u> (between the means) is due to chance / is not significant

Question	Answer	Marks	Guidance
2(b)	1 measure / AW, <u>time</u> spent, feeding / eating / AW, on each food source ; any one from: 2 <i>idea of</i> method to identify, individual / song type, (male) <i>G. conirostris</i> ; 3 <i>idea of</i> method to observe (feeding of male) <i>G. conirostris</i> on each food source ;	2	allow bird / finch as AW for <i>G. conirostris</i> ignore repeats / do a statistical test each food source = prickly pear and palo santo tree (or several different sources) 1 allow note / record / calculate as AW for measure allow 'how long' as AW for time ignore observe time for mp1 ignore time at location (needs to be time spent eating) ignore calculate, percentage / proportion (of time) INTERNAL MAX 2 allow use of, rings / paint / markings to identify, individual / song type ignore listen unqualified 3 allow lab-based or field-based methods do not accept ref. to birds with only one food source (birds must have a choice of food sources) A natural habitat (with both food sources) and observe A feeding stations with both food sources and observe A cage / AW, containing both food sources and observe A camera / AW (to observe), birds (feeding) on both food sources

Question	Answer	Marks	Guidance
2(c)	any four from: <i>supports the conclusion:</i> 1 <i>idea that</i> different groups (of <i>G. conirostris</i>) would have different selection pressures ; 2 <i>idea that</i> differences between the two groups (of <i>G. conirostris</i>) will lead to, reproductive / genetic / behavioural / ecological, isolation ; 3 there is no geographical separation ; ora <i>extend the investigation:</i> 4 study female (<i>G. conirostris</i>) ;	4	allow bird / finch as AW for <i>G. conirostris</i> allow beak as AW for bill ignore <i>ref. to</i> different islands / allopatric speciation 1 & 2 allow population / types as AW for groups R <i>ref. to</i> (two) species (penalise once and ECF) 1 different / two, groups = song type A and song type B / different bill lengths ignore (changing) allele frequencies unqualified allow description of selection pressure: e.g. A (male) <i>G. conirostris</i> with different bill lengths are <u>adapted</u> to feed on different food sources e.g. A female <i>G. conirostris</i> will <u>select one</u> song type of male <i>G. conirostris</i> for <u>mating</u> 2 & 3 allow separation as AW for isolation 3 A ora same geographical location A no physical barrier A geographical ranges (of the two groups) overlap ignore improvements to the original investigation ignore repeat investigation unqualified ignore repeat, in different seasons / at different times of the year ignore study a larger sample ignore do a statistical test

Question	Answer	Marks	Guidance
2(c)	5 <i>idea of studying (G. conirostris) for several generations ;</i> 6 use DNA analysis / AW ;		5 ignore longer time unqualified allow many / multiple as AW for several allow several years if stated, allow 3 or more years as AW for several years 6 allow DNA sequencing / DNA profiling / genetic fingerprinting / genetic screening